



Aircraft Detection, Classification & Segmentation for Air Traffic Control

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An end-to-end computer-vision pipeline combining pretrained YOLOv8 detection & segmentation with a custom-trained ResNet-50 classifier to localise, segment and identify aircraft types from runway imagery.

GROUP MEMBERS

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ABSTRACT

Modern air-traffic monitoring needs reliable, automated interpretation of visual feeds. **ATC-Vision** performs three vision tasks on one image: **detection** (locating every aircraft), **segmentation** (pixel-level instance masks) and **classification** (identifying the exact aircraft type). Detection and segmentation reuse a pretrained YOLOv8m-seg backbone, while a ResNet-50 fine-tuned on eight aircraft classes reaches **98.97%** test accuracy. A hybrid stage routes every detected box through the classifier, so generic “airplane” detections become named types with confidence and runway position.

OBJECTIVES

- Detect and localise all aircraft in a runway / apron scene.
- Produce instance segmentation masks for situational awareness.
- Classify each aircraft into one of 8 types with confidence.
- Group types into Commercial / Military / Training categories.
- Deliver results through an interactive analysis dashboard.

SYSTEM WORKFLOW



Hybrid design — YOLOv8 answers *where & how many*; the fine-tuned ResNet-50 answers *which type*. A single-aircraft image can skip detection and go straight to the classifier.

DETECTION

YOLOv8m-seg

Single-stage detector predicting boxes for the COCO airplane class at a 0.35 confidence threshold; boxes are clipped to image bounds and passed downstream for type identification.

SEGMENTATION

Instance Masks

The same network outputs per-instance masks, visualised step-by-step: input → preprocessing → binary masks → overlay → contours, with per-region coverage statistics.

CLASSIFICATION

ResNet-50 (Fine-tuned)

ImageNet-pretrained ResNet-50 with a new 8-way head, 224×224 input and ImageNet normalisation; returns the top class, confidence and full probability distribution.

MODEL & DATASET

Backbone	ResNet-50	Classifier head	Linear 2048→8	Input size	224 × 224
Detector / Segmenter	YOLOv8m-seg	Test images	678	Inference / image	576 ms
ATR Airbus Boeing C130 F16 Grob KAI Sukhoi 8 classes · Commercial / Military / Training					

99.0%

ACCURACY

99.0%

PRECISION (WEIGHTED)

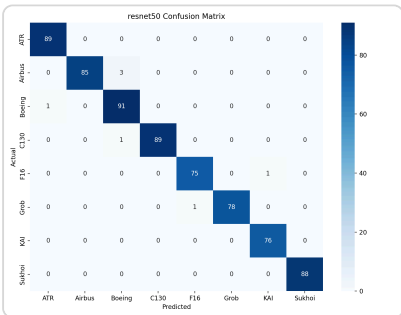
99.0%

RECALL (WEIGHTED)

99.0%

F1-SCORE (WEIGHTED)

CONFUSION MATRIX



Strongly diagonal — errors are rare and confined to visually similar commercial jets.

PER-CLASS ACCURACY

ATR	100.0%
Airbus	96.6%
Boeing	98.9%
C130	98.9%
F16	98.7%
Grob	98.7%
KAI	100.0%
Sukhoi	100.0%

CONCLUSION & FUTURE WORK

The hybrid YOLOv8 + ResNet-50 design delivers accurate, explainable aircraft identification (98.97% test accuracy) while reusing strong pretrained detectors, keeping the trainable footprint small. Future work: fine-tune the detector on aircraft-specific data, extend to real-time video, and add range / altitude estimation.

TECHNOLOGY STACK

PyTorch · torchvision · Ultralytics YOLOv8
OpenCV · Flask · React + Vite